CLASS 34 – MODELLING TEXT

1. **Unsupervised Methods and Topic Modelling**
2. **Latent Dirichlet Allocation (LDA) Topic Modelling**
3. **Key Takeaways**
4. **Appendix I – Supervised Methods and Keyword Counting**

Sources: Chen, Rubin, and Cornwall (2021); Müller and Guido (2016)

1. UNSUPERVISED METHODS AND TOPIC MODELLING

UNSUPERVISED LEARNING

The problem set this week has you experiment with some supervised text analysis approaches to predict speaker identity. We can also take an unsupervised approach to modelling text.

- Unsupervised methods simultaneously estimate a set of categories and assign documents to these categories without reference to previously provided categories

- Cluster analysis is an example of this, and you could apply K-Means or another clustering algorithm to the document term matrix

- It is convenient to refer to the clusters as **"topics"**. However, simple models are only conducting word associations[1]

[1] Techniques exist to try to provide key words to help in identifying the topics. See for example Eshima, Shusei, Kosuke Imai, and Tomoya Sasaki. "Keyword-assisted topic models." American Journal of Political Science 68.2 (2024): 730-750.

Documents can be assigned together for several reasons based on which words they have in common:

- The words could be content driven
- The words could be indicative of author style, but have little independent meaning

After conducting a cluster analysis, we can review the words that generated cluster assignments and the document assignments themselves to see if they provide meaningful context.

QUESTION: What are some policy questions where it may be useful to have texts grouped into different topics?

DEFINING TOPICS

Read the following paragraphs. Fill in the final column of the table to record what you think the topic of the text could be.

Quote ID	Text	Topic
1	On Monday, in one gruelling practice run after another, she defends herself against her teammates as they replicate the bullying tactics she may experience as world No 1 in Paris. Rival paddles and boats bash against her but Woods surges ahead, negotiates the heavy buoys, flips through an Eskimo roll under water, flies across the rapids, and finishes the course in a minute's worth of power and skill.	
2	John Foreman, the UK's former defence attaché to Moscow, said Scholz's reluctance to send Taurus missiles to Kyiv was another example of how the German chancellor was "following, not leading" when it came to the war in Ukraine.	
3	…one of her earliest memories was of an apparent drug raid at a neighbouring house: "I remember looking out the window and saw a Swat team rushing in. It's a bit crazy how I lived in a neighbourhood like that but when I go home now it's way different.	
4	On the pitch, Lionel Messi led Argentina to glory in a final that ranks among the greatest games of all time. There were more goals scored (172) than at any previous World Cup. According to FIFA, 3.4 million people attended the games, and the four-year cycle to 2022 was the most lucrative ever, generating $7.5 billion.	
5	Of that wider figure of 15,021, thousands of deaths have been attributed to cardiac arrest or natural causes. Human Rights Watch, Amnesty International and several other campaign groups are handling thousands of cases in which families of dead migrant workers believe the cause of death has been deliberately misclassified to avoid any acknowledgement of wrongdoing and resulting compensation.	

SINGLE MEMBERSHIP VS. MIXED MEMBERSHIP MODELS

In the above table, some of the quotes are drawn from the same newspaper articles:

- Quotes 1 and 3 appear in a Guardian article about Olympic hopeful Kimberley Woods.
    - The story focuses on Woods as an athlete but also on her personal history of bullying and self-harm
- Quotes 4 and 5 appear in an ESPN article reviewing the 2022 FIFA World Cup
    - A main focus on the article is on criticisms and responses to allegations of human rights violations in Qatar (and other countries) when preparing for the world cup, but several paragraphs present more factual information about the results of the competition

Different portions of texts may fall into separate categories. Not all models are well suited to classify documents into **multiple categories**, but trying to force every document into a single category reduces the available information.

Single membership models will partition all documents into a single category, with no overlap.

- K-Means as we have used it is an example of a single membership model

- While simple, these models may fail to capture nuance, especially in longer documents that move between several topics

Mixed membership models can allocate documents into more than one category.

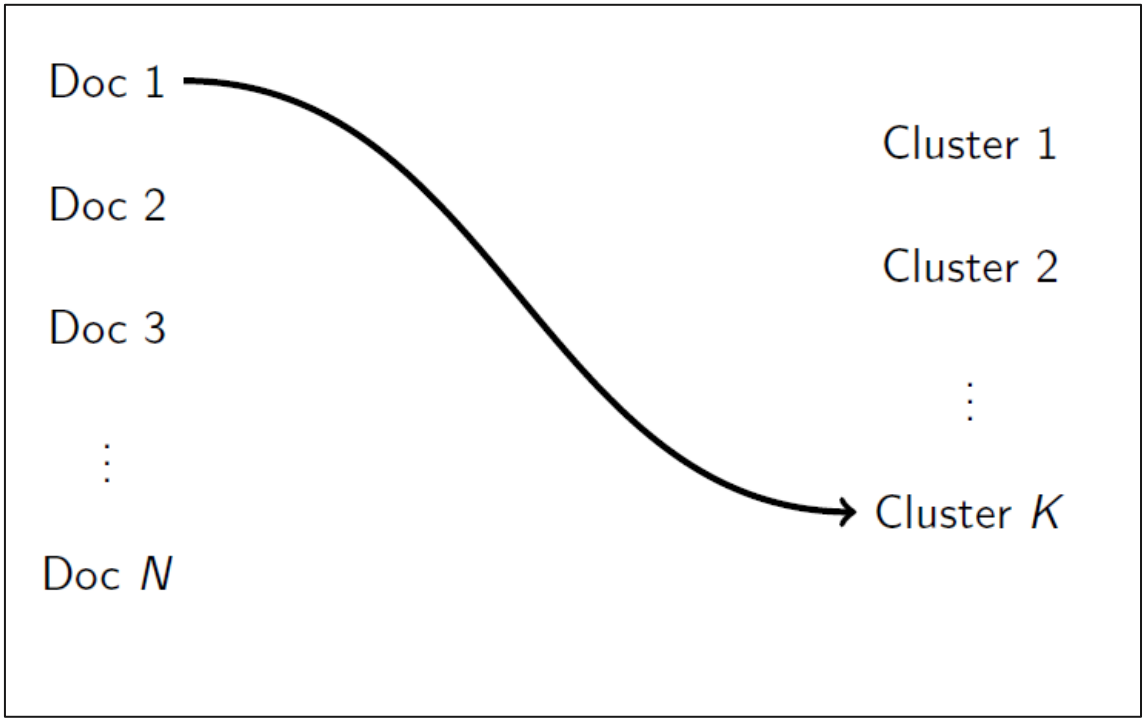
- We can assign a document to be comprised of several topics according to some a probability function.

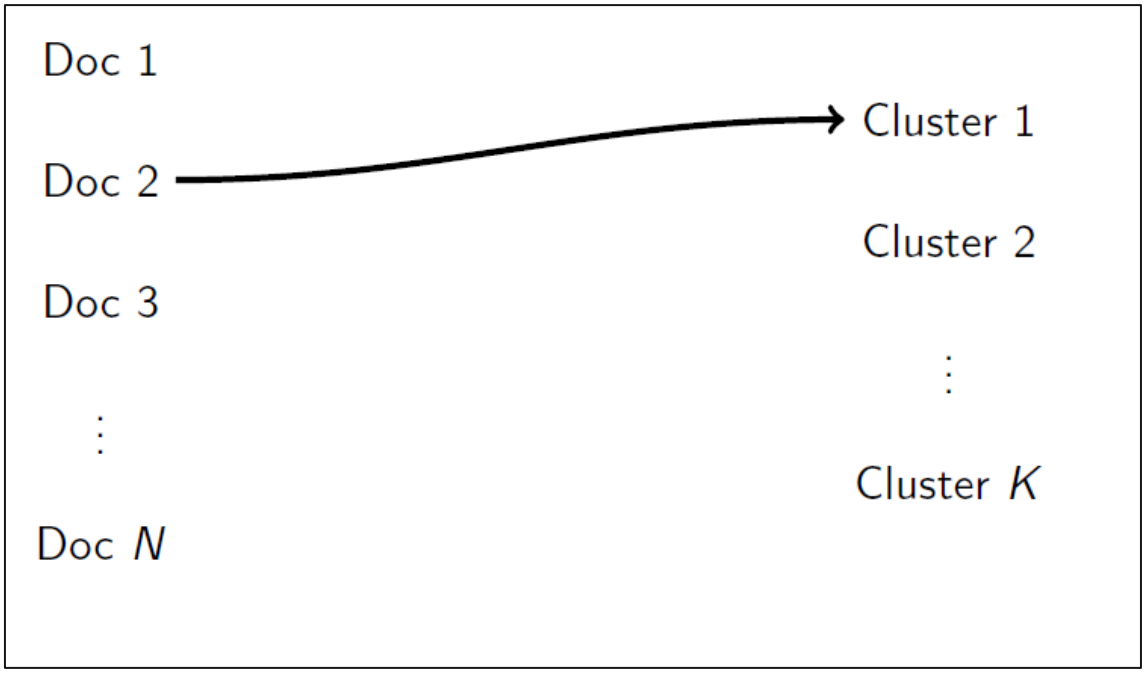
- This will be a probability distribution <u>at the document level</u> over which topic it belongs to.

    - The article on Kimberly Woods above could be classified as 60% sports and 40% self-care

Single Membership Models:

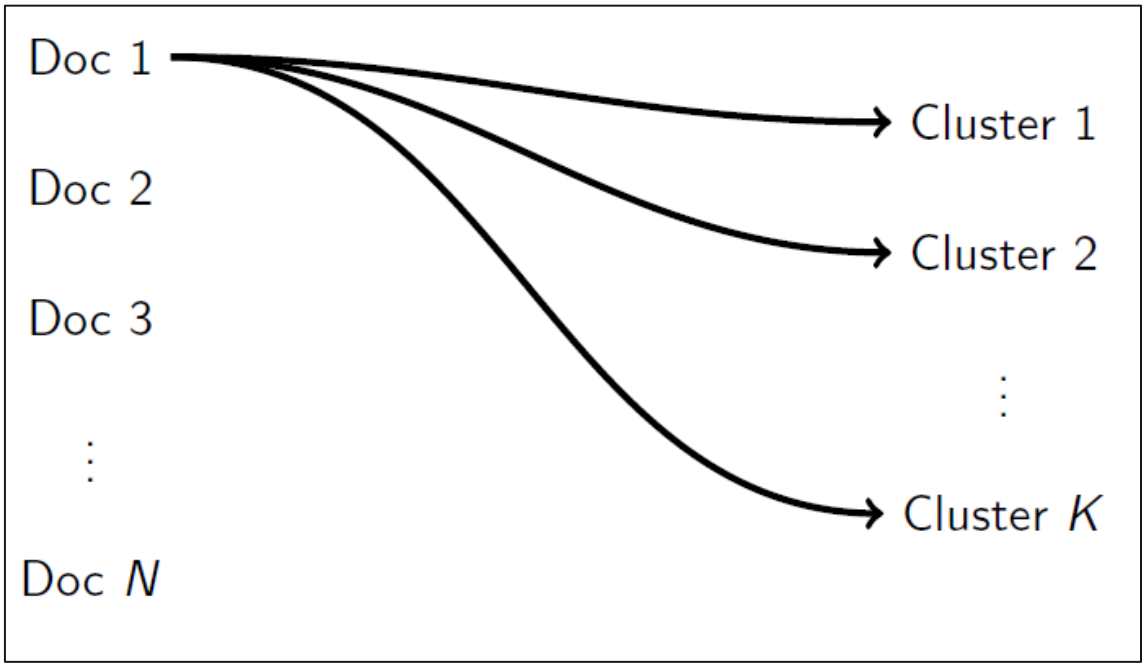
Each document is matched to <u>one and only</u> one cluster (topic)





Mixed Membership Models:

Individual documents are assigned to multiple clusters (topics)



In this case the arrows in the above plot could represent different probability weightings for Document 1 across the relevant clusters.

The probability weights will not in general be equal. A document will be more heavily associated with a smaller number of clusters (topics).

2. LATENT DIRICHLET ALLOCATION (LDA) TOPIC MODELLING

Latent Dirichlet Allocation (LDA) is a popular **mixed membership** model for assigning documents to different topics.

LDA is an unsupervised model which will group documents into different clusters (which we will refer to as "topics" in this context).

There will be three different core matrices for this model.

1. The document term matrix (DTM) **[INPUT]**
    - Our bag of words representation used to fit the model
2. The "topic-term" matrix **[OUTPUT]**
    - Associations of individual words (tokens) probabilistically with different topics (e.g. connecting the token "ball" to the topic "sports")
3. The "document-topic" matrix **[OUTPUT]**
    - Associations of individual documents probabilistically with different topics (e.g. assigning a paragraph to be about "sports")
    - Documents can represent a mixture of topics (e.g. 70% sports, 30% human rights).

This model begins with a pre-defined number of topics you wish to group documents into (equivalent to setting the value of *k* in the K-Means algorithm).

Individual words can appear in multiple topics. As an example, consider newspaper articles that could fall into topics like *Financial News* or *Movie Reviews*.

- The word "bond" may be a frequent in *Financial News*
- The word "bond" could also appear in *Movie Reviews* for a James Bond film
- However, we may think the relative frequency of the word "bond" is higher in the *Financial News* category than the *Movie Reviews* category

LDA MOTIVATING EXAMPLE[2]

Let's motivate topic assignment in a way a human might. In this example, we will make use of a smaller corpus and human knowledge to outline the principles a computer may apply to this problem.

Consider the following example sentences (perhaps from a larger corpus):

1. I like to eat broccoli and bananas.
2. I ate a banana and spinach smoothie for breakfast.
3. Hamsters and kittens are cute.
4. My sister adopted a kitten yesterday.
5. Look at this cute hamster munching on a piece of broccoli.

Next, let's apply topic modelling to allocate these sentences into different groups. First, let's begin by assuming there are 2 topics. Our goal will be to identify these topics based on word **co-occurrence**.

A human may come up with the following classifications.

Topic A: (Food)

Topic B: (Pets)

Which could be applied to the above sentences with word associations.

1. I like to eat broccoli and bananas.
2. I ate a banana and spinach smoothie for breakfast yesterday.
3. Hamsters and kittens are cute.
4. My sister adopted a kitten yesterday.
5. Look at this cute hamster munching on a piece of broccoli.

Some words can be associated with foods / eating, others are associated with pets.

[2] This example builds upon materially originally presented by Adeline Lo

LDA OUTPUT 1: TOPIC-TERM MATRIX

1. I like to eat broccoli and bananas.
2. I ate a banana and spinach smoothie for breakfast yesterday.
3. Hamsters and kittens are cute.
4. My sister adopted a kitten yesterday.
5. Look at this cute hamster munching on a piece of broccoli.

Conditional #1: <u>If</u> we have documents assigned to different topics, we can calculate relative probabilities of different words appearing in different topics.

The below table shows possible output if we ran this process over many documents (not just the sentences above). To save space, only a subset of words is shown.

Topic	broccoli	banana	breakfast	Kitten	cute	hamster	like	yesterday	**Total**
A	0.3	0.25	0.2	0.01	0.01	0.01	0.12	0.1	100%
B	0.01	0.01	0.01	0.35	0.24	0.25	0.08	0.05	100%

The entry of 0.3 for Topic A and word broccoli indicates that if a word is being selected from "Topic A", there is a 30% chance the word will be "broccoli".

- The row sums of this matrix will be 100%.

Some words will be more strongly associated with individual topics.

- "broccoli" has a higher probability of being drawn from Topic A than Topic B
- "banana" may not be the most likely word for Topic A, but it is more likely to be selected from Topic A than Topic B
    - If we see the word "banana" in a document, this will make it more likely the document is from Topic A

The model will associate a small probability for every word with every topic. Without this, the model may frequently multiply probabilities by zero when encountering new words, which would make the assignments unstable.

LDA OUTPUT 2: DOCUMENT-TOPIC MATRIX

1. I like to eat broccoli and bananas.
2. I ate a banana and spinach smoothie for breakfast yesterday.
3. Hamsters and kittens are cute.
4. My sister adopted a kitten yesterday.
5. Look at this cute hamster munching on a piece of broccoli.

Conditional #2: <u>If</u> we have the relative probabilities of words within each topic, we can use these probabilities to assign different documents to different topics.

The below table shows a potential example for the above sentences:

Document	Topic A Weight	Topic B Weight	Total
1	0.99	0.01	100%
2	0.99	0.01	100%
3	0.01	0.99	100%
4	0.01	0.99	100%
5	0.60	0.40	100%

The "Topic A Weight" column reports the estimated probability a document belongs to a specific topic. Probabilities will sum to 100% across the topics (the columns).

Based on the word probabilities, Documents 1, 2, 3 and 4 are heavily weighted towards singular topics. Topic 5 is a mixed topic as it discusses pets and eating.

Similar to the topic-term matrix, the document-topic matrix will associate a small probability for every document with every probability. Again, this helps to avoid instability that can be caused from multiplying by zero.

In the above examples, we relied on two conditional statements to make our calculations.

1. If we have documents assigned to different topics, we can calculate relative probabilities of different words appearing in different topics.

2. If we have the relative probabilities of words within each topic, we can use these probabilities to assign different documents to different topics.

QUESTION: When starting the problem, which conditional statement do we know?

Of course, when we start the model, computers have access to **neither** document-topic assignments **nor** topic-term assignments.

However, this problem is the same issue presented in the K-Means algorithm. In the K-Means context, we could say:

1. If observations are assigned to a cluster, we can calculate cluster centroids

2. If we have cluster centroids, we can assign observations to the closest cluster

We used iteration to solve K-Means, and LDA will apply the same idea in its approach.[3] LDA will iteratively assign documents to topics and update the word associations and topic associations.

[3] The actual algorithm used will be different for LDA, with many implementations relying on the EM algorithm and more assumptions on underlying probability distributions.

SAMPLE TOPIC MODEL SAMPLE OUTPUT

Grimmer (2012)[4] applied topic modelling to politicians' press releases to examine how politicians vary in the extent they use these releases to communicate their opinions on different issues or claim credit for policies. [5]

After running a topic model with k=44 different topics, press releases were grouped into different topics. The following table presents three of these topics, along with the discriminating words (words most used to assign a document to that topic).

Computer Topic Label	Discriminating Words
1	iraq, iraqi, troop, war, sectarian
2	honor, prayer, remember, fund, tribute
3	firefight, homeland, afgp, award, equipment

This basic LDA model can provide these discriminating words, but it can only provide uninformative labels of the topics (e.g. Topic 1, Topic 2, *etc.*).

[4] Grimmer, Justin, Solomon Messing, and Sean J. Westwood. "How words and money cultivate a personal vote: The effect of legislator credit claiming on constituent credit allocation." *American Political Science Review* 106.4 (2012): 703-719.
[5] As reported in Grimmer and Stewart (2013), Table 4

A human may then interpret and label the above topics as:

Human Coded Topic Label	**Discriminating Words**
Iraq War	iraq, iraqi, troop, war, sectarian
Honorary	honor, prayer, remember, fund, tribute
Fire Department Grants	firefight, homeland, afgp, award, equipment

You should also then read a sample of documents assigned to a given topic to validate how well the topic grouping has been undertaken.

This process still requires analyst effort to label the topics, but the effort comes in interpreting the results after the model has been fit, rather than in trying to generate this information upfront.

COSTS AND BENEFITS OF TOPIC MODELLING

Like other text analysis techniques, topic modelling will not generally outperform human classification and understanding of documents. The bag-of-words and other assumptions of the model are vastly simplifying language.

There are however several benefits of these and other computational methods.

- **Scale** – once constructed, a topic model can be applied to thousands of instances of text in a matter of seconds. It would be much more expensive and time consuming to try to have humans read and classify all of these texts.

- **Consistency** – computational methods are consistent. After the model is fit, we can evaluate the exact relationship between words, documents, and topics.

    a. A computer will consistently apply this coding scheme to all documents.

    b. Human coders may be prone to bias (both conscious and unconscious).

- **New insights** - The model may identify new areas analysts were not considering.

Further, the model as presented has no pre-conceptions about what words mean, it only examines relative co-occurrences of words within the data you are providing.

- Contrast this to the vector representations of words presented in the guest lecture
- These representations try to extract meaning on individual words from their historical training set, so the models will propagate any biases in that training data

3. KEY TAKEAWAYS

- After creating a DTM, we can apply many of the supervised and unsupervised techniques already introduced to find patterns between the bag-of-words representation and an outcome of interest.
    - This may still require appropriate scaling, especially if the documents are not the same length
- Topic modelling is a completely unsupervised technique for clustering groups of topics together based on word similarity.
    - Documents are assumed to be composed of different topics
    - Different topics have different relative word frequencies
    - The text of the document itself is determined by first selecting amongst its topics and then selecting amongst the words associated with a topic
- Computational text analysis delivers speed, scalability, and consistency in coding beyond human performance, but this comes at a cost of missing nuances in text that humans can more easily identify
- Dictionary and keyword counting methods are a fast way to perform initial analysis, especially if existing dictionaries are reliable (see appendix)

APPENDIX I – SUPERVISED METHODS AND KEYWORD COUNTING

EXAMPLES OF SUPERVISED LEARNING

We can use the DTM as we would any other data frame. If we have a label or outcome associated with each document, we could apply supervised learning models to examine the relationships between word frequency and these outcomes.

Examples of Supervised Learning Problems in Text Analysis

- Identifying speakers / authors from word choice
- Identify political parties from floor speeches
- Categorize documents as generally positive or negative in tone
- Identify patterns in which texts face censorship[6]
    - The assigned reading for this week examines this and you should plan to read this for an upcoming seminar discussion
- Track how politicians engage in credit claiming

These methods rely on labels, which can be produced by hand coding a subset of documents.

Once these labels are generated, we can create training and testing data sets and build a model.

[6] King, Gary, Jennifer Pan, and Margaret E. Roberts. "How censorship in China allows government criticism but silences collective expression." *American political science Review* 107.2 (2013): 326-343.

KEYWORD COUNTING AND DICTIONARY METHODS

Relatedly, we could engage in **keyword counting**. This type of analysis counts the instances of particular words or phrases.

- You could see how frequently the word "inflation" appears in government press releases and track this over time.

Dictionary methods are a special case of keyword counting. They look at weighted averages of predefined keywords to classify documents into categories.

A simple example could involve classifying the tone (or sentiment) of a document as being positive or negative. To do this:

- Identify individual words as having a positive or negative tone
- Assign a score to the sentiment of these words. The simplest weighting would assign a score of -1 to negative words, and +1 to positive words, with neutral words getting a score of zero
- Add up the relative counts of negative and positive words within a document to arrive at an overall sentiment score

This scoring could be extended to any classification you wish to study; however, you may need to generate an appropriate dictionary if it does not already exist.

- Broad sentiment (Positive vs. Negative)
- Specific emotions (Anger, Joy, Sadness)
- Non-sentiment dictionaries (*e.g.,* counts of words associated with food or travel)

SENTIMENT LEXICONS

There are many existing dictionaries for this type of sentiment analysis. A few examples are:

- *Bing Lexicon* – for binary classification[7]
- *National Research Council (NRC) Emotion Lexicon*
    - English word associations for eight categories (anger, fear, anticipation, true, surprise, sadness, joy and disgust)
- *AFINN* dictionary, which scores word sentiments from -5 to 5
- Affective Norms for English Words (ANEW) – a dictionary specifically focused on studies of focus and attention
- Linguistic Inquire and Word Count (LIWC) – an available paid service with over 100 language dimensions

General guidance for dictionary methods:

- Pre-existing dictionaries are convenient, but if you have the time, creating your own dictionary is well worth it.
- There are also automated methods to assist in identifying useful words to include in a dictionary if some documents already have been assigned to different categories.[8]

The words used to construct dictionaries in this manner can be problem specific. The goal is to find words that can separate different categories of interest and then measure how often those words appear in text.

[7] Liu, Bing, Minqing Hu, and Junsheng Cheng. "Opinion observer: analyzing and comparing opinions on the web." *Proceedings of the 14th international conference on World Wide Web*. 2005.
[8] Monroe, Burt L., Michael P. Colaresi, and Kevin M. Quinn. "Fightin'words: Lexical feature selection and evaluation for identifying the content of political conflict." *Political Analysis* 16.4 (2008): 372-403.

DICTIONARY METHODS VS. SUPERVISED LEARNING

Supervised learning is an attractive alternative to dictionary methods because it does not require us to identify important words beforehand.

We can allow our model to learn how words relate to the classification we provide in the initial set of labelled documents.

Supervised learning is therefore domain specific by design.

- If you fit a supervised learning model to a set of tax documents, it will identify the important discriminating words in this context automatically.

A key drawback of supervised learning is that we must be careful to ensure our model is not overfitting. This requires having enough hand-coded documents in the training set to accurately learn patterns.

- For basic tasks, hundreds to thousands of documents will need to be hand labelled, and a consistent coding scheme will need to be developed for this

- For more complicated modelling, or for training sets with more tokens, more labelled documents will be needed